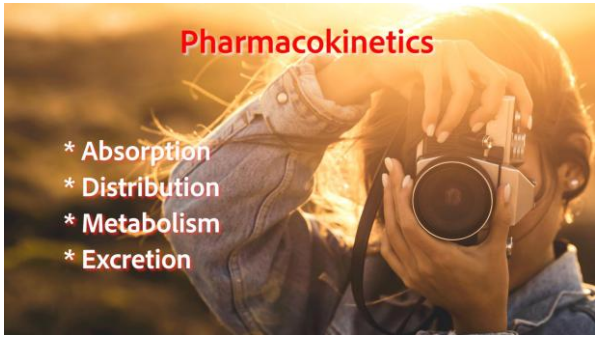
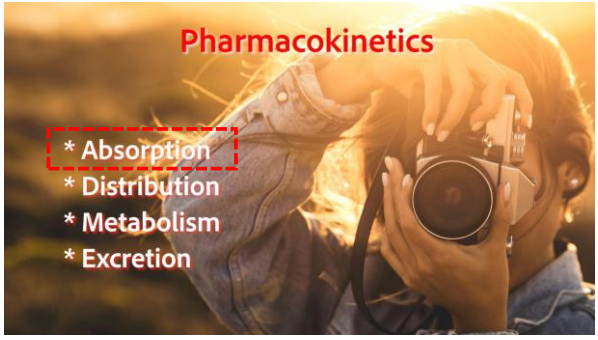
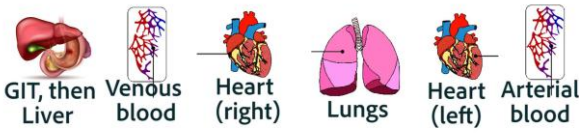


1



2

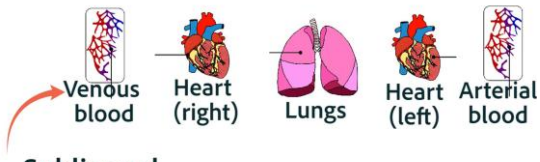


Medication absorbed from the GIT will need to travel this entire distance after absorption.

This takes time!



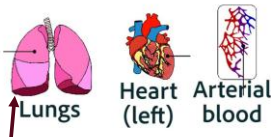
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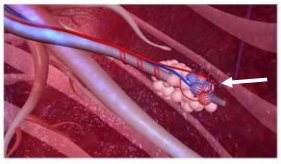


Sublingual
Transdermal patch
Suppository (IN LOWER RECTUM)

4

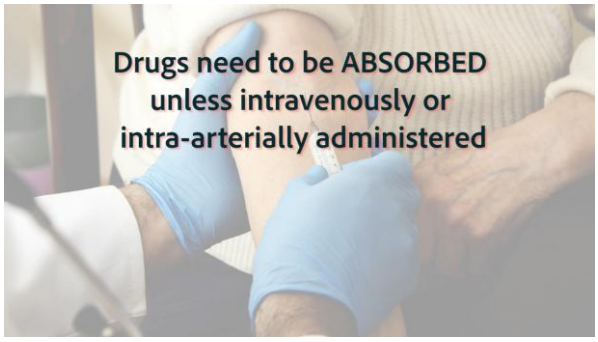
Inhalation results in **rapid absorption**





Accesses blood here

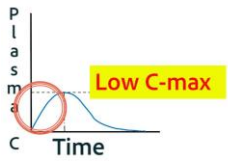
5



6

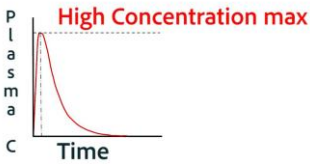


Oral
Slowly absorbed





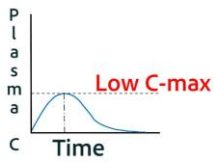
Inhalation
Rapidly absorbed



7

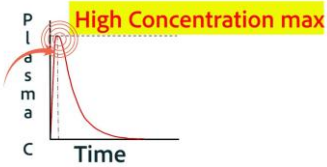


Oral
Slowly absorbed





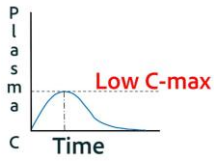
Inhalation
Rapidly absorbed



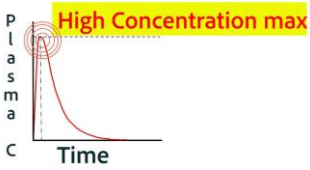
8

Which of the following would have the greatest potential for side effects?

Oral
Slowly absorbed

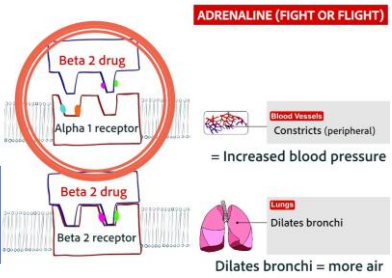


Inhalation
Rapidly absorbed



9

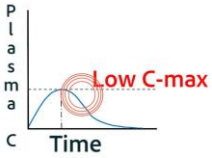
A beta 2 drug may have *some* binding to other receptors, especially in high concentrations



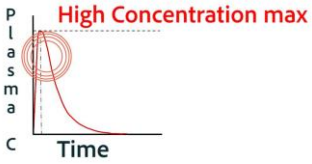
10

Which of the following has the fastest metabolism in the first instance?

Oral
Slowly absorbed



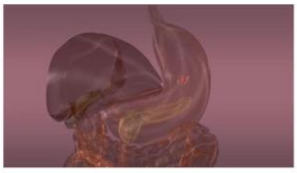
Inhalation
Rapidly absorbed



11

Absorption of oral medications

Why is there very little absorption in stomach?

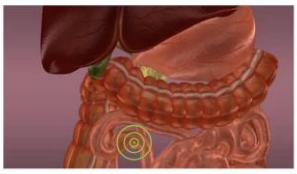


1. Drugs spend little time in stomach
2. There is a small area available for absorption

12

Absorption of oral medications

Most occurs in the small intestine



13

Factors involved in absorption from GIT

Factors specfic to the drug

* Molecular size
* Acidic or alkaline properties
* Polarity (lipophilic or hydrophilic)

Example: Fat soluble (lipophilic) drugs are absorbed faster

14

Factors involved in absorption from GIT

Factors specific to the GIT at the time

* Motility of the GIT: high motility = less absorption
* Blood flow to the area (higher blood flow increases absorption rate)

15

What effect does food have on absorption of drugs?

* Increases absorption RATE
* Decreases BIOAVAILABILITY



16

Generally, when is it best to take oral medications?

One hour before meals

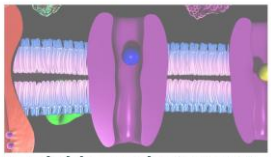
OR

Two hours after meals

17

Mechanisms of absorption

Simple diffusion (most common)



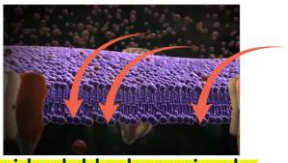
Water soluble products must use protein channels in cell surface



18

Mechanisms of absorption

Simple diffusion (most common)



Lipid soluble drugs simply diffuse THROUGH the cell

19

The best absorption of medications comes with:

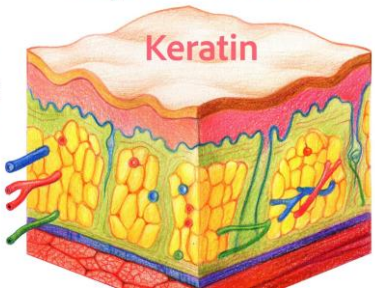
a. Lipophilic (fat 'loving'; fat soluble) drugs

b. Hydrophilic (water 'loving'; water soluble)

20

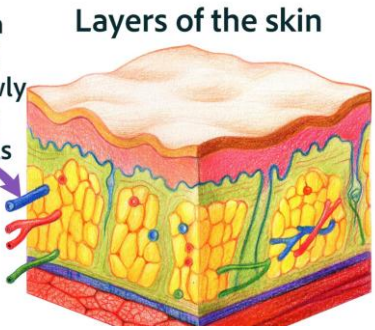
Layers of the skin

Topical creams generally stay on the top layers

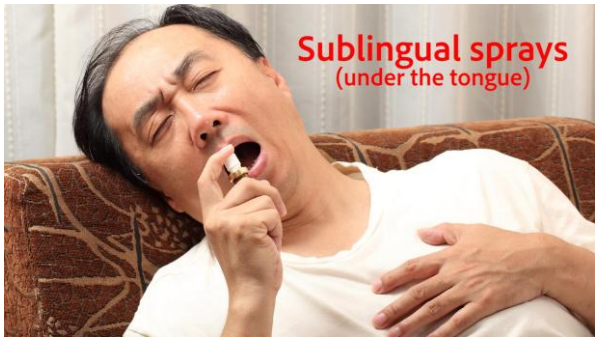


21

Drugs within transdermal patches slowly permeate to blood vessels



22



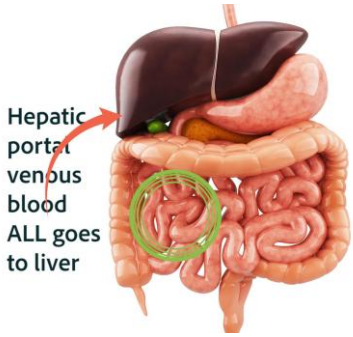
23

Always use medications in the way in which they are recommended.

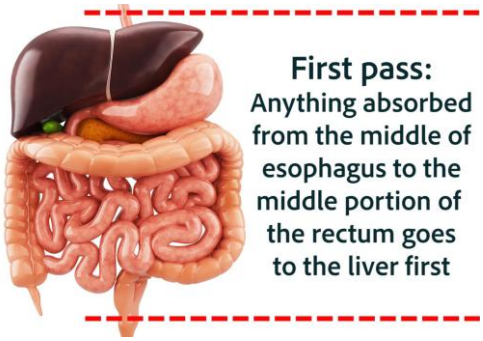
24


Pharmacokinetics
* Absorption
* Distribution
* Metabolism
* Excretion

25

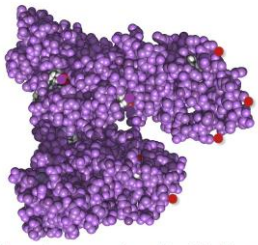

Hepatic portal venous blood ALL goes to liver

26


First pass:
Anything absorbed from the middle of esophagus to the middle portion of the rectum goes to the liver first

27

Example: 80% bound to proteins 20% free



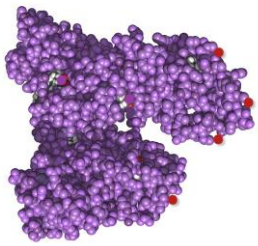
Albumin complexed with drug

Only this unbound portion is active !



28

80% Bound



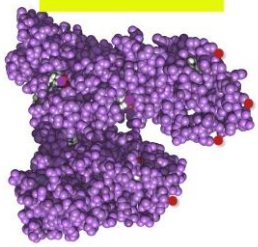
20% Free

Free drug goes into cells



29

100% Bound

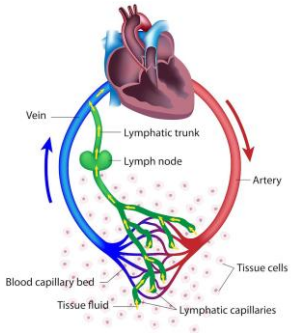


0% Free

When this happens, an equal amount is released from the protein binding to establish an equilibrium.



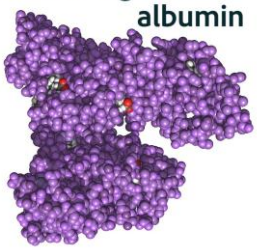
30



Drugs establish an equilibrium between body compartments

31

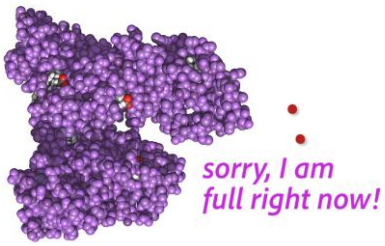
Acidic drugs bind to albumin



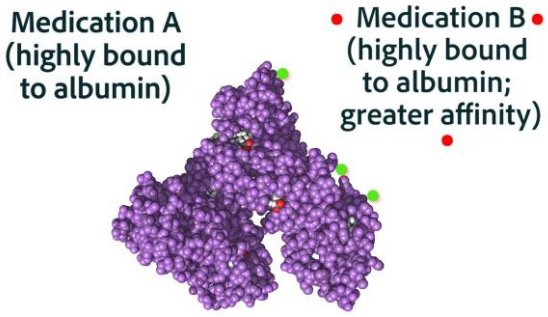
Alkaline drugs bind to globulins

...but there are limits to the amount carried

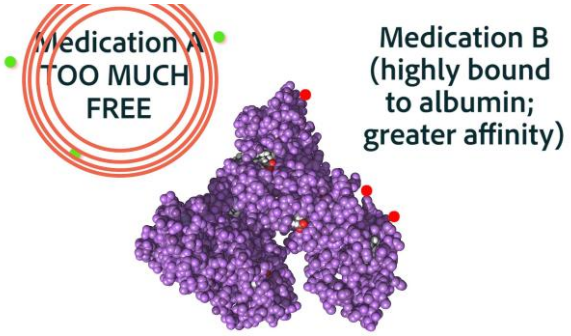
32



33



34



35

Conditions of decreased proteins

* Liver disease (the liver makes many proteins)

* Kidney disease (glomerular disorders lose proteins)

* Inadequate intake

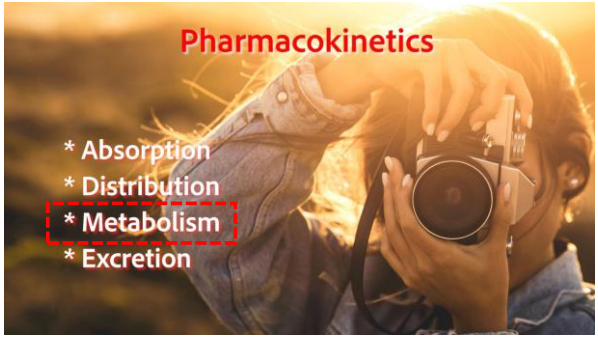
36

...speaking of the liver...



...next up, metabolism of drugs

37



38

Some drugs are extensively metabolized in the hepatic 'FIRST PASS'

Glycerol trinitrate (nitroglycerine) for angina has a low bioavailability (less than 1%)

Less than 1% gets into circulation unchanged

Morphine for pain has a low bioavailability (30%)



39

Some drugs are extensively metabolized in the hepatic 'FIRST PASS'



Morphine bioavailability:

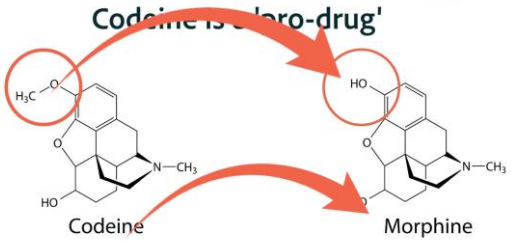
30% orally

100% by IV

40

Administering oral 'pro-drugs'

Codeine is a pro-drug'



41

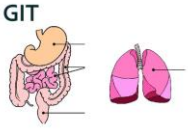
Sites of drug metabolism

Primary site:
The liver



* Has the enzymes necessary
* Receives blood from entire GIT

The GIT and other organs also metabolize drugs



42

The hepatic 'first pass':

What does it do to the drug? It can:

* Make an active drug into an inactive drug
* Make an inactive drug into an active drug
* Make an active drug into a more active drug
* Make the drug more toxic (as it does with ethanol)

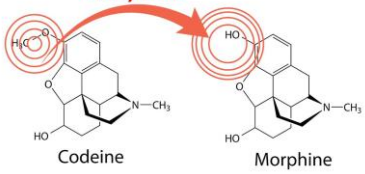
The primary goal of liver metabolism is to make the drug water soluble and easier to excrete

43

Metabolism of drugs in the liver

Phase 1 reactions

Cytochrome P450 enzymes



Codeine Morphine

44

Metabolism of drugs in the liver

Phase 1 reactions

Cytochrome P450 enzymes

This polar group makes the drug more hydrophilic



Codeine Morphine

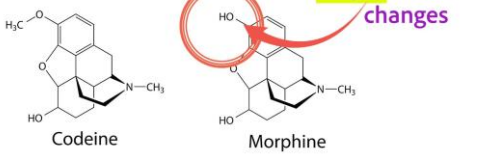
45

Metabolism of drugs in the liver

Phase 1 reactions

Cytochrome P450 enzymes

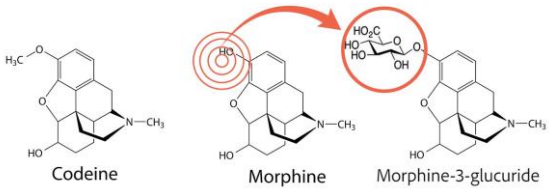
This can act as a target for further changes



46

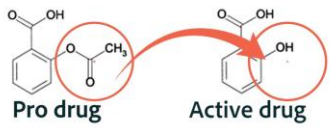
Metabolism of drugs in the liver

Phase 2 reactions



47

Phase 1 reaction

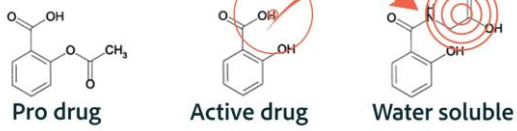


Metabolism of Aspirin



48

Phase 1 reaction Phase 2 reaction



Pro drug Active drug Water soluble

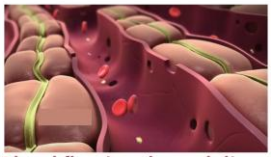
Metabolism of Aspirin



49

On the first pass through the liver the drug is made:

a. Active
b. More active
c. Less active
d. Inert
e. More toxic
f. Less toxic
g. More water soluble
h. All of the above



Blood flowing through liver

50

On the first pass through the liver the drug is made:

a. Active
b. More active
c. Less active
d. Inert
e. More toxic
f. Less toxic
g. More water soluble
h. All of the above

The liver is trying to make the drug water soluble to make it easier to excrete. The process may inadvertently make the drug more toxic.

Ethanol → Acetaldehyde

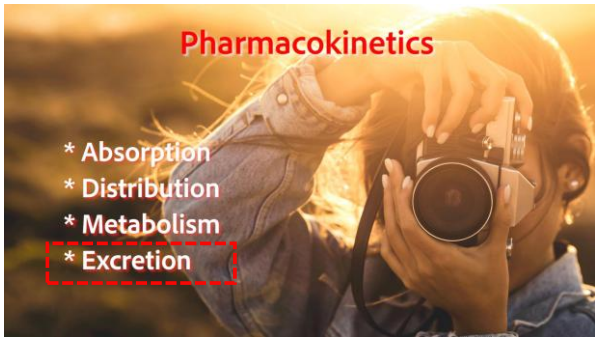
More toxic

51

Next up:

The excretion of medications...

52



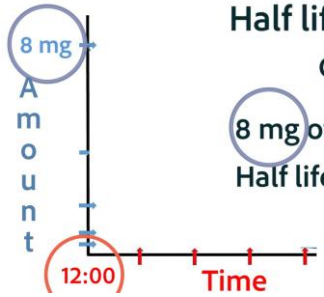
53

Half life

The time that it takes to reduce the concentration of the drug to half of its original amount



54



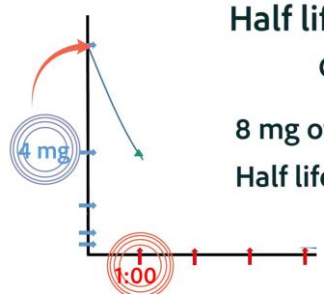
Half life

Consider this:

8 mg of the drug at 12:00

Half life of one hour

55



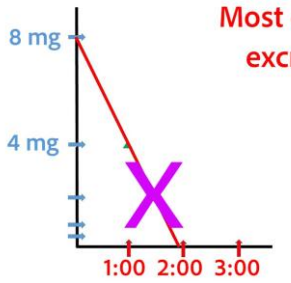
Half life

Consider this:

8 mg of the drug at 12:00

Half life of one hour

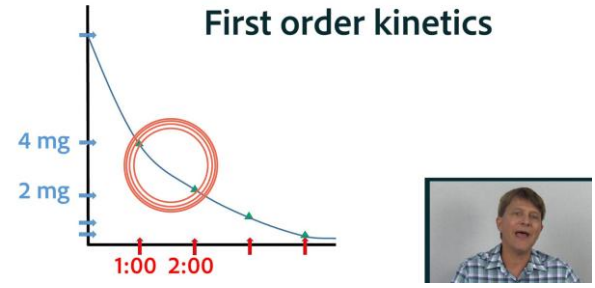
56



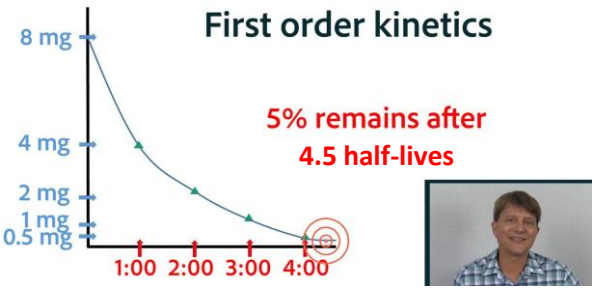
Most drugs don't get excreted like this



57



58



59

Half life is relevant to drugs metabolized by first order kinetics

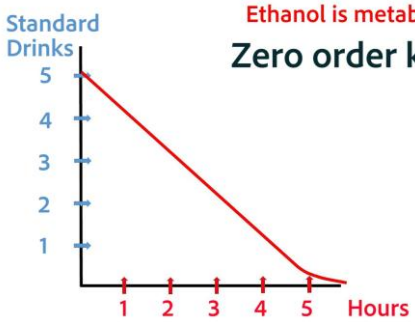
60

Half life is relevant to drugs metabolized by first order kinetics

Most drugs are metabolized by first order kinetics

Ethanol is not

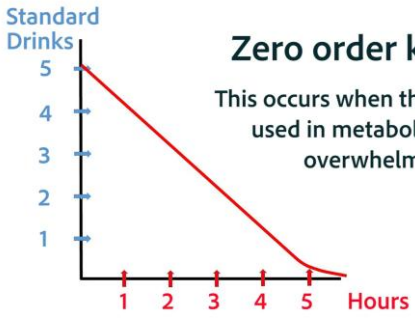
61



Ethanol is metabolized by

Zero order kinetics

62

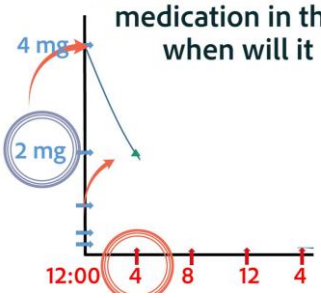


Zero order kinetics

This occurs when the enzymes used in metabolism are overwhelmed

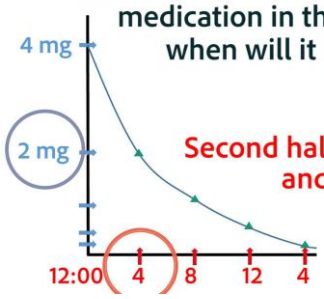
63

Half life of 4 hours: If there is 4 mg of medication in the system at noon, when will it be down to 1 mg?



64

Half life of 4 hours: If there is 4 mg of medication in the system at noon, when will it be down to 1 mg?



Second half life takes 4 hours and starts out at 2 mg

65



66

Drug class

A group of meds that have a similar structure or work by the same mechanism

Antacids neutralize the stomach acid

Histamine 2 receptor blockers reduce acid by blocking the histamine 2 receptor

Cimetidine

Famotidine

'tidine' suffix

67

Beta blockers

Slow down heart; decrease blood pressure

Metoprolol, Atenolol, Propranolol, Nebivolol, Timolol

Look for the 'olol' suffix

5 of the top 200 most prescribed

Benzodiazepines

Anti anxiety; anti-seizure; CNS depressant

Alprazolam, Clonazepam, Lorazepam, Diazepam, Temazepam

'azepam' or 'azolam' suffix

5 of the top 200 most prescribed

Corticosteroids

Powerful immune system suppressants

Fluticasone, Hydrocortisone, Mometasone, Prednisolone, Methylprednisolone

'asone'; 'pred'; 'cort' affixes (and others)

8 of the top 200 most prescribed

H 2 Receptor blocker

Decreases stomach acid

Famotidine, Cimetidine, Nizatidine

Look for the 'tidine' suffix

4 of the top 200 most prescribed

68

Histamine 2 Blockers

Decrease stomach acid by blocking the Histamine 2 receptor in the parietal cells of the stomach

- Cimetidine
- Famotidine



Indicated in the treatment of PUD and GORD/GERD

69

What are the responsibilities of the stomach acid?

* Begins to degrade foodstuffs
* Helps in the absorption of nutrients
* Decreases microorganisms
* Helps in the absorption of drugs

70



What would some of the side effects be as you greatly reduce stomach acid?

71



Stomach acid:

Helps nutrient absorption

Helps decrease bacteria

Helps absorb drugs

What happens if we decrease all of that?

72



73

Vitamin B12

nutrient deficiencies

calcium

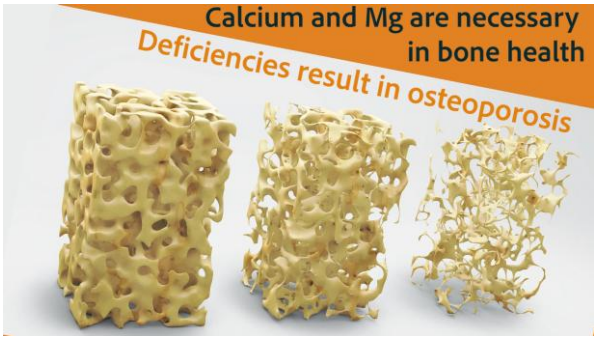
magnesium

74

Vitamin B12 deficiency in people taking histamine 2 blockers can result in a form of dementia!



75



76

Other side effects of the H2 blockers

* Increased GIT AND respiratory tract infections
* Decreased absorption of some drugs
* Increased incidence of food allergies
* Possible dysrhythmias



77

Histamine 2 Blockers

Decrease stomach acid by blocking the Histamine 2 receptor in the parietal cells of the stomach

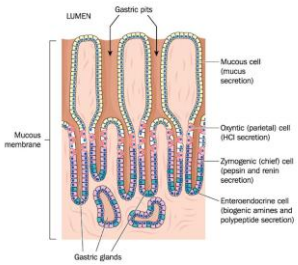
- **Cimetidine**
- **Famotidine**



Indicated in the treatment of PUD and GORD/GERD

78

Histamine 2 Receptor Blockers



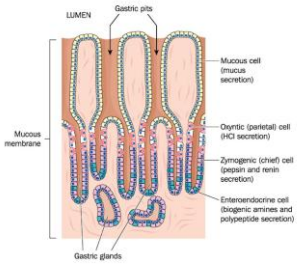
H+



* Histamine
* Acetylcholine
* Gastrin

79

Histamine 2 Receptor Blockers





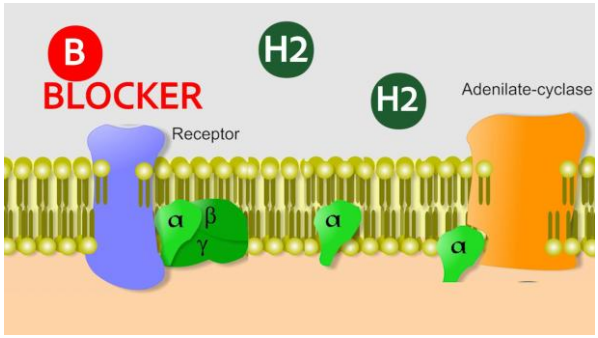
Histamine 2 antagonists block SOME of the stimuli for HCl release

80

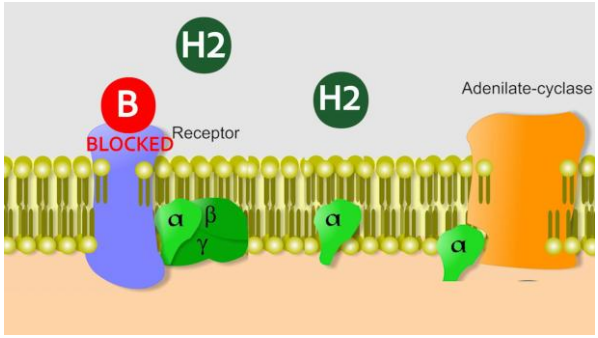
Histamine 2 antagonists reduce acid by up to 80-90%

Proton pump inhibitors reduce acid by up to 99-100%

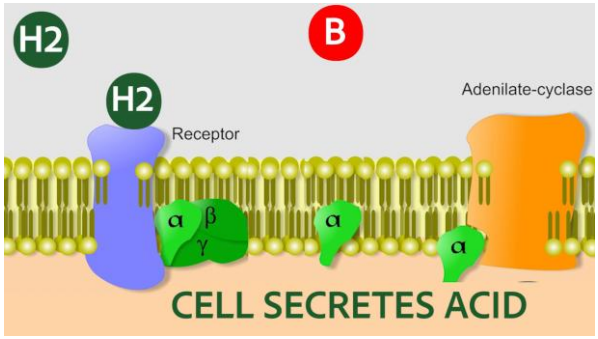
81


B
BLOCKER
H2
H2
Adenilate-cyclase
Receptor
α
β
γ
α
α

82


H2
B
BLOCKED
H2
Adenilate-cyclase
Receptor
α
β
γ
α
α

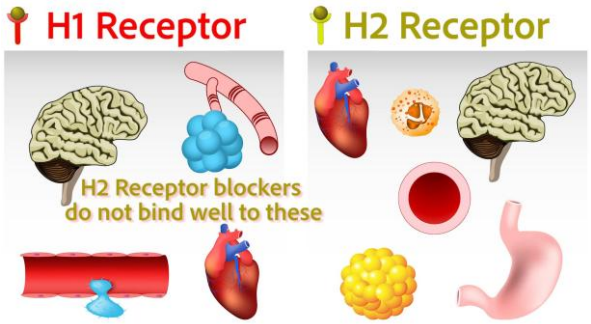
83


H2
B
H2
Adenilate-cyclase
Receptor
α
β
γ
α
α
CELL SECRETES ACID

84

Is the Histamine 2 blocker an antihistamine?

85



86

Antihistamines (H1 antagonists) are fairly selective for the H1 receptor (although some block acetylcholine receptors)

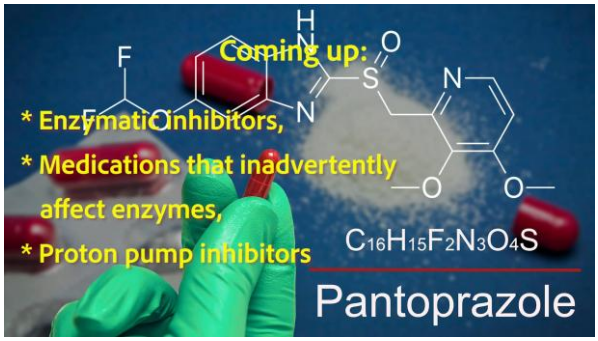
87

Chapter summary

Main topic: Pharmacokinetics

* Absorption
* Distribution
* Metabolism
* Excretion

88



89

References

- Dr Ken Wojcikowski
- PharmacoPhoto
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90